



COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

Quiz # 2

Course Title:	VLSI Design	Course Code:	CPE434	Credit Hours:	4(3,1)
Course Instructor:	Dr. Muhammad Naeem Awais	Programme Name:	BS Computer Engineering		
Semester:		Batch:		Date:	7 th April 2026
Time Allowed:	10 Minutes		Maximum Marks:	5	
Student's Name			Registration. No.:		

Question # 1:

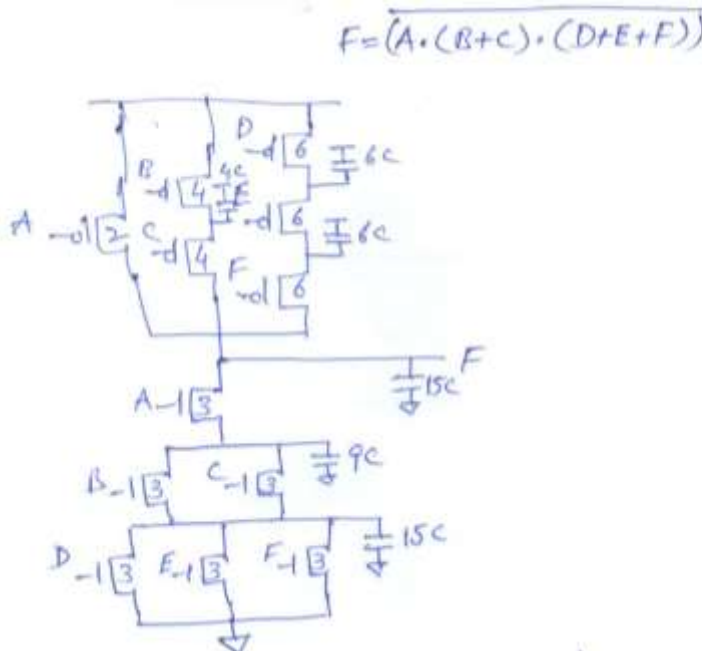
(PLO2-CLO2-C4)

For the compound logic implementation of the following gate:

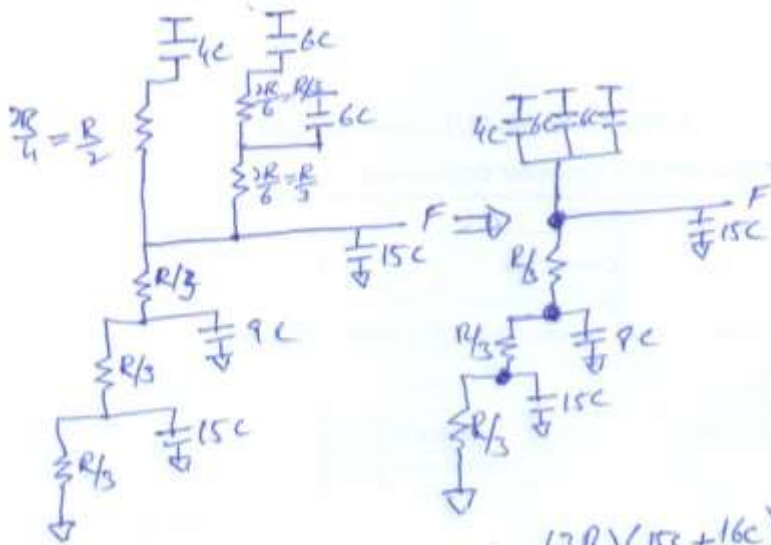
$$F = \overline{(A \cdot (B+C) \cdot (D+E+F))}$$

- Compute the transistor widths chosen to achieve effective rise and fall resistance equal to that of a unit inverter (R).
- Analyze the circuit to estimate the worst-case falling delay for the following input combination:
 $A=1, B=1, C=0, D=1, E=0, F=0$
- Analyze the circuit to estimate the worst-case rising delay for the following input combination:
 $A=1, B=1, C=1, D=0, E=0, F=0$

Solution:



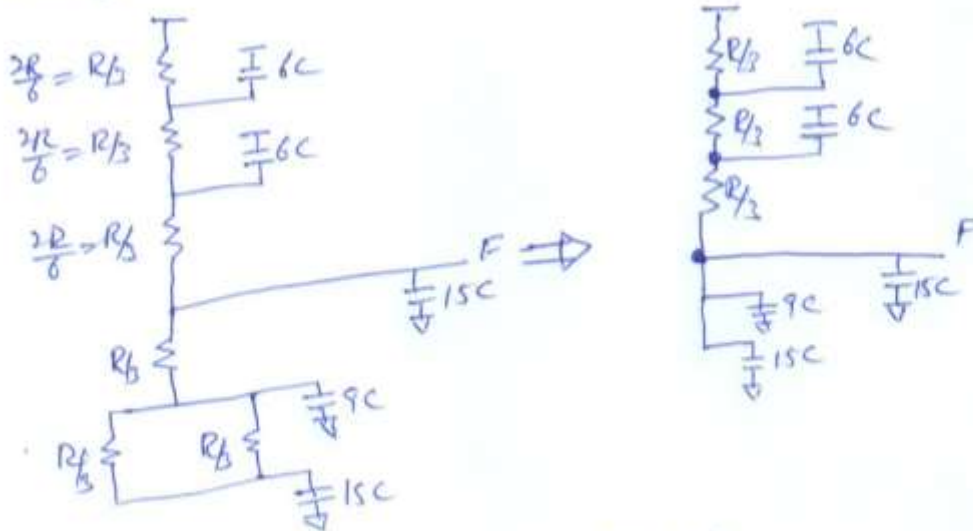
Falling Delay: (A=1, B=1, C=0, D=1, E=0, F=0)



$$t_{pd_f} = \left(\frac{R}{3}\right)(15C) + \left(\frac{2R}{3}\right)(6C) + \left(\frac{3R}{3}\right)(15C + 16C)$$

$$= 5RC + 6RC + 31RC = 42RC$$

Rising Delay: (A=1, B=1, C=1, D=0, E=0, F=0)



$$t_{pd_r} = \left(\frac{R}{3}\right)(6C) + \left(\frac{2R}{3}\right)(6C) + \left(\frac{3R}{3}\right)(15C + 24C)$$

$$= 2RC + 4RC + 39RC = 45RC$$